

DEVOPS PROJECT

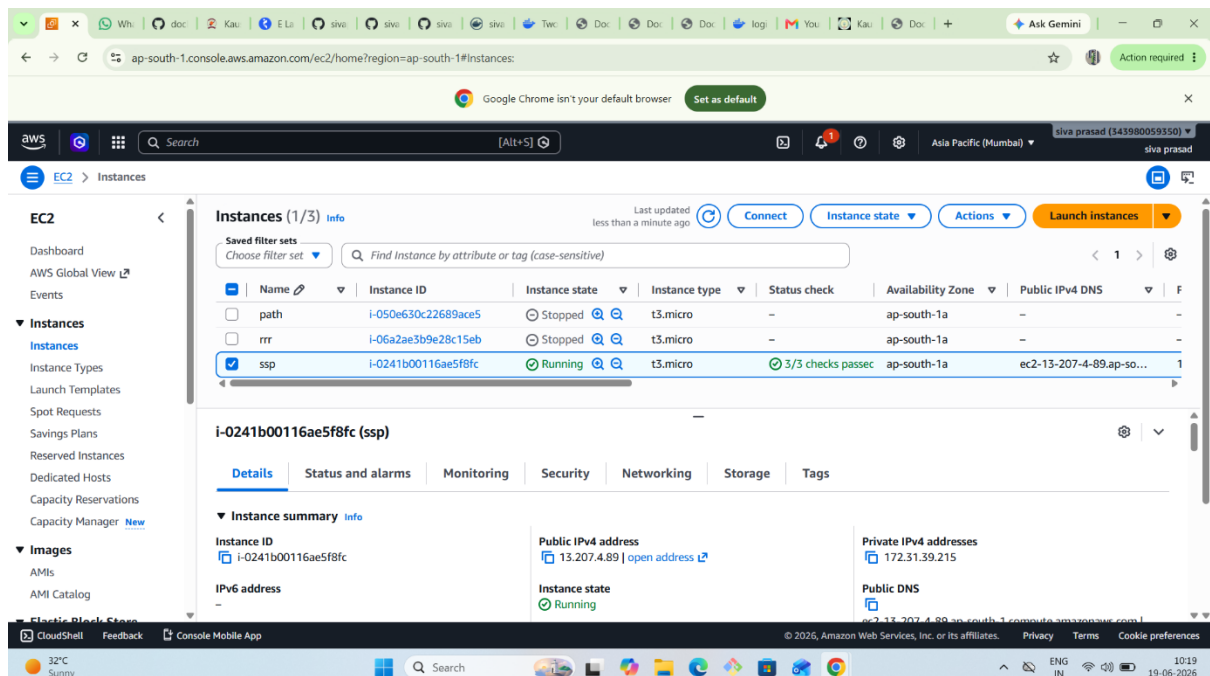
Task 1 - Server Setup and SSH Configuration

🚩 Objective

Provision a Linux server and configure secure SSH access using key-based authentication.

🖥️ Environment

* OS: Ubuntu (Virtual Machine)



⚙️ Steps Performed

1. Update System

```
MINGW32/c/Users/Others/downloads/docker-project/task-1
$ cd Downloads
bash: cd: Downloads: No such file or directory

Others@V13AVKIMAB MINGW32 ~/Desktop
$ ssh -i "asp.pem" ubuntu@ec2-13-207-4-89.ap-south-1.compute.amazonaws.com
The authenticity of host 'ec2-13-207-4-89.ap-south-1.compute.amazonaws.com (13.207.4.89)' can't be established.
ED25519 key fingerprint is SHA256:Zg977kTv9pouJhTvTmUshJIEERP-SPDf50DFWUNDM.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-13-207-4-89.ap-south-1.compute.amazonaws.com' (ED25519) to the list of known hosts.
welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Thu Jun 18 15:57:49 UTC 2026

System load:  0.03      Temperature:  -273.1 C
Usage of /:   30.6% of 6.61GB   Processes:    119
Memory usage: 27%           Users logged in: 0
Swap usage:   0%             IPv4 address for ens5: 172.31.39.215

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/*copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ubuntu@ip-172-31-39-215:~$ ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/ubuntu/.ssh/id_ed25519):
Enter passphrase for '/home/ubuntu/.ssh/id_ed25519' (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/ubuntu/.ssh/id_ed25519
Your public key has been saved in /home/ubuntu/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:SAVij6kF7uPj0nrj8dAVQdPqbanH126V0avNW5M ubuntu@ip-172-31-39-215
The key's randomart image is:
+--[ED25519 256]--+
|  . . . . .
|
+-----+
X
```

sudo apt update

sudo apt upgrade -y

2. Install SSH

sudo apt install openssh-server -y

sudo systemctl start ssh

sudo systemctl enable ssh

3. Check SSH Status

sudo systemctl status ssh

```
ubuntu@ip-172-31-46-138: ~
Welcome to Ubuntu 20.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Jun 20 01:39:44 UTC 2026

System load:  0.0          Temperature: -273.1 C
Usage of /:   33.8% of 6.61GB   Processes:    117
Memory usage: 27%           Users logged in: 0
Swap usage:   0%             IPv4 address for ens5: 172.31.46.138

Expanded Security Maintenance for Applications is not enabled.

17 updates can be applied immediately.
14 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Sat Jun 20 01:27:57 2026 from 106.192.17.90
ubuntu@ip-172-31-46-138:~$ sudo -i
root@ip-172-31-46-138:~# sudo apt update
sudo apt install openssh-server -y
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
17 packages can be upgraded. Run 'apt list --upgradable' to see them.
openssh-server is already the newest version (1:10.2p1-2ubuntu3.2).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 17
root@ip-172-31-46-138:~# sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Drop-In: /usr/lib/systemd/system/ssh.service.d
            └─ec2-instance-connect.conf
   Active: active (running) since Sat 2026-06-20 01:22:56 UTC; 18min ago
   Invocation: #f818960145ac22ba4f1559ec1d0233
   TriggeredBy: ● ssh.socket
   Docs: man:sshd(8)
          man:sshd_config(5)
   Main PID: 1017 (sshd)
   Tasks: 1 (limit: 627)
   Memory: 6.8M (peak: 8M)
   CPU: 102ms
   CGroup: /system.slice/ssh.service
           └─1017 "sshd: /usr/sbin/sshd -D -o AuthorizedKeysCommand /usr/shar

Jun 20 01:22:56 ip-172-31-46-138 systemd[1]: Starting ssh.service - OpenBSD Secure Shell server:
Jun 20 01:22:56 ip-172-31-46-138 sshd[1017]: Server listening on 0.0.0.0 port 22.
Jun 20 01:22:56 ip-172-31-46-138 systemd[1]: Started ssh.service - OpenBSD Secure Shell server:
Jun 20 01:22:56 ip-172-31-46-138 sshd[1017]: Server listening on :: port 22.
```

4. Find IP Address

ip a

5. Connect via SSH (Password)

ssh username@server_ip

6. Generate SSH Key

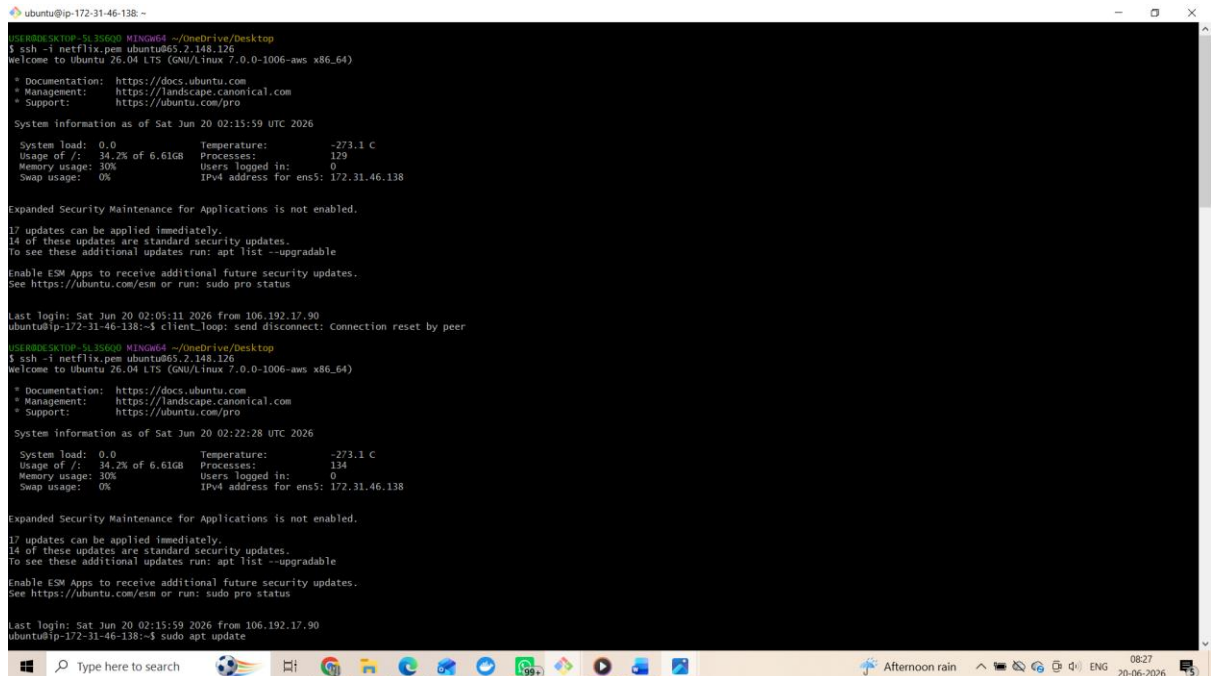
ssh-keygen

7. Copy Key to Server

ssh-copy-id username@server_ip

8. Login Without Password

ssh username@server_ip



The screenshot shows a Windows terminal window titled 'ubuntu@ip-172-31-46-138: ~'. The user runs the command `ssh -i netflix.pem ubuntu@65.2.148.126`. The terminal output shows the SSH connection process, including the system information of the remote Ubuntu 26.04 LTS server. The output indicates that the connection was successful and that the user is logged in as 'ubuntu'. The terminal also shows the system information of the remote server, including the system load, usage of /, memory usage, swap usage, temperature, processes, users logged in, and the IPv4 address for ens5. The terminal window has a taskbar at the bottom with various application icons and a system tray showing the date and time as 08:27 on 20-06-2026.

```
ubuntu@ip-172-31-46-138: ~  
USER@DESKTOP-SL35GQD MINGW64 ~/OneDrive/Desktop  
$ ssh -i netflix.pem ubuntu@65.2.148.126  
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)  
  
* Documentation:  https://docs.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:       https://ubuntu.com/pro  
  
System information as of Sat Jun 20 02:15:59 UTC 2026  
System load:  0.0          Temperature: -273.1 C  
Usage of /:   34.2% of 6.61GB Processes:      129  
Memory usage: 30%         Users logged in: 0  
Swap usage:   0%          IPv4 address for ens5: 172.31.46.138  
  
Expanded Security Maintenance for Applications is not enabled.  
17 updates can be applied immediately.  
14 of these updates are standard security updates.  
To see these additional updates run: apt list --upgradable  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
Last login: Sat Jun 20 02:05:11 2026 from 106.192.17.90  
ubuntu@ip-172-31-46-138:~$ client_loop: send disconnect: Connection reset by peer  
USER@DESKTOP-SL35GQD MINGW64 ~/OneDrive/Desktop  
$ ssh -i netflix.pem ubuntu@65.2.148.126  
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)  
  
* Documentation:  https://docs.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:       https://ubuntu.com/pro  
  
System information as of Sat Jun 20 02:22:28 UTC 2026  
System load:  0.0          Temperature: -273.1 C  
Usage of /:   34.2% of 6.61GB Processes:      134  
Memory usage: 30%         Users logged in: 0  
Swap usage:   0%          IPv4 address for ens5: 172.31.46.138  
  
Expanded Security Maintenance for Applications is not enabled.  
17 updates can be applied immediately.  
14 of these updates are standard security updates.  
To see these additional updates run: apt list --upgradable  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
Last login: Sat Jun 20 02:15:59 2026 from 106.192.17.90  
ubuntu@ip-172-31-46-138:~$ sudo apt update
```

9. Disable Password Login (Security)

sudo nano /etc/ssh/sshd_config

Change:

PasswordAuthentication no

Restart:

```
sudo systemctl restart ssh
```

🛡️ Security Features

- * SSH enabled
- * Key-based authentication configured
- * Password login disabled

📁 Files Used

- * ~/.ssh/id_rsa
- * ~/.ssh/id_rsa.pub
- * /etc/ssh/sshd_config

✅ Expected Outcome

Secure SSH login without password using SSH keys.

Task 2 - DOCKER CONTAINER DEPLOYMENT

🚩 Objective

Learn basic Docker commands and container usage.

🔧 Steps Performed

1. Checked Docker version:

```
docker --version
```

2. Pulled nginx image:

```
docker pull nginx
```

3. Ran container:

```
docker run -d -p 8080:80 nginx
```

4. Checked running containers:

```
docker ps
```

📷 Output

```
MINGW32/c/Users/Others/downloads/docker-project/task-1
client_loop: send disconnect: Connection reset by peer

Others@V1JAYKIMAM MINGW32 ~/downloads
$ ssh -i "ssp.pem" ubuntu@ec2-13-207-4-89.ap-south-1.compute.amazonaws.com
Welcome to Ubuntu 20.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Jun 18 16:56:04 UTC 2026

System load: 0.0          Temperature: -273.1 C
Usage of /:  38.2% of 6.61GB   Processes:    119
Memory usage: 33%          Users logged in: 0
Swap usage:  0%             IPv4 address for ens5: 172.31.39.215

 * Ubuntu Pro delivers the most comprehensive open source security and
   compliance features.
   https://ubuntu.com/aws/pro

Expanded Security Maintenance for Applications is not enabled.

14 updates can be applied immediately.
14 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Thu Jun 18 16:28:19 2026 from 27.59.60.149
ubuntu@ip-172-31-39-215:~$ sudo -i
root@ip-172-31-39-215:~# docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu4.1
root@ip-172-31-39-215:~# mkdir docker-app
cd docker-app
root@ip-172-31-39-215:~/docker-app# nano index.html
index.html
root@ip-172-31-39-215:~/docker-app# nano Dockerfile
root@ip-172-31-39-215:~/docker-app# docker build -t siva .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  4.096kB
Step 1/2 : FROM nginx:latest
latest: Pulling from library/nginx
c82e83b01f69: Pulling fs layer
8da80f8205ea: Pulling fs layer
93181f19640f: Pulling fs layer
3461bb128618: Pulling fs layer
868d78dcead: Pulling fs layer
ac3e3151b8c0: Pulling fs layer
```

📌 Conclusion

Understood how to use Docker to run containers.

Task 3 - CONTAINER MONITORING LOGS

📌 Objective

Create a custom Docker image using a Dockerfile.

🔧 Steps Performed

Now I will show container logs with timestamps.

Command:

`docker logs <container_id>`

OR

`bash monitor.sh`

This displays logs along with timestamps, which helps in monitoring container activity

```
MINGW32/c/Users/Others/downloads/docker-project/Task-1
create mode 100644 task1/README.md
create mode 100644 task1/Dockerfile
create mode 100644 task1/README.md
create mode 100644 task1/index.html
create mode 100644 task1/README.md
create mode 100644 task1/README.md
root@ip-172-31-39-215:~/docker-project# git remote add origin https://github.com/sival70/docker-project.git
git branch -M main
git push -u origin main
Username for 'https://github.com': sival70
Password for 'https://sival70@github.com': client_loop: send disconnect: Connection reset by peer

Others@V13AKUMAR MINGW32 ~/downloads
$ ssh -i "asp.pem" ubuntu@ec2-13-207-4-89.ap-south-1.compute.amazonaws.com
Welcome to Ubuntu 20.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Fri Jun 19 01:46:43 UTC 2026
System load: 0.0          Temperature: -273.1 C
Usage of /: 41.8% of 6.61GB Processes: 129
Memory usage: 36%        Users logged in: 0
Swap usage: 0%           IPv4 address for ens5: 172.31.39.215

 * Ubuntu Pro delivers the most comprehensive open source security and
  compliance features.
  https://ubuntu.com/awp/pro

Expanded Security Maintenance for Applications is not enabled.

14 updates can be applied immediately.
14 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Thu Jun 18 18:36:56 2026 from 106.221.176.244
ubuntu@ip-172-31-39-215:~$ sudo -i
expected one of these actions: --help, --version, --remove-timestamp, --validate, --list, --edit, --login, --shell, a command as a positional argument, --reset-timestamp
usage: sudo -h | -k | -k | -V
usage: sudo [-ABKNS] [-p prompt] [-D directory] [-g group] [-u user] [-i | -s] [command [arg ...]]
usage: sudo -i [-ABKNS] [-p prompt] [-g group] [-u user] [command [arg ...]]
usage: sudo -s [-ABKNS] [-p prompt] [-g group] [-u user] [command [arg ...]]
usage: sudo -e [-ABKNS] [-p prompt] [-D directory] [-g group] [-u user] file ...
ubuntu@ip-172-31-39-215:~$ sudo -i
root@ip-172-31-39-215:~# sudo mkdir -p /opt/container-monitor/logs
root@ip-172-31-39-215:~# nano /opt/container-monitor/monitor.sh
root@ip-172-31-39-215:~# chmod +x /opt/container-monitor/monitor.sh
root@ip-172-31-39-215:~# /opt/container-monitor/monitor.sh
root@ip-172-31-39-215:~# cat /opt/container-monitor/logs/monitor.log
```

```
MINGW32/c/Users/Others/downloads/docker-project/Task-1
Running kernel seems to be up-to-date.
No services need to be restarted.
No containers need to be restarted.
No user sessions are running outdated binaries.
No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-39-215:~$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu4.1
ubuntu@ip-172-31-39-215:~$ top
top - 18:39:25 up 43 min, 2 users, load averages: 0.00, 0.04, 0.01
Tasks: 121 total, 1 running, 120 sleeping, 0 stopped, 0 zombie
Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
Mem Mem : 908.7 total, 159.2 free, 363.2 used, 501.9 buff/cache
Mem Swap: 0.0 total, 0.0 free, 0.0 used, 545.5 avail Mem

  PID USER      PR  NI    TID    RES     SHR     S    CPU    MEM    TIME COMMAND
2047 root        20   0 1793236 46352 32244 S    0.3   5.0   0:00.58 containerd
2554 ubuntu    20   0 11036   5760 3572 R    0.3   0.6   0:00.19 top
1 root        20   0 25520 16260 10968 S    0.0   1.7   0:03.46 systemd
2 root        20   0 0 0 0 0 S    0.0   0.0   0:00.00 kthreadd
3 root        20   0 0 0 0 0 S    0.0   0.0   0:00.00 pool_workqueue_release
4 root        0 -20 0 0 0 0 I    0.0   0.0   0:00.00 kworker/R-cpu_gg
5 root        0 -20 0 0 0 0 I    0.0   0.0   0:00.00 kworker/R-sync_wq
6 root        0 -20 0 0 0 0 I    0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
7 root        0 -20 0 0 0 0 I    0.0   0.0   0:00.00 kworker/R-slab_flushwq
8 root        0 -20 0 0 0 0 I    0.0   0.0   0:00.00 kworker/R-netns
10 root       0 -20 0 0 0 0 I    0.0   0.0   0:00.00 kworker/0:0H-kblockd
13 root       0 -20 0 0 0 0 I    0.0   0.0   0:00.00 kworker/R-mm_percpu_wq
14 root       20   0 0 0 0 0 S    0.0   0.0   0:00.00 ksoftirqd/0
15 root       20   0 0 0 0 0 I    0.0   0.0   0:00.10 rcu_sched
16 root       20   0 0 0 0 0 S    0.0   0.0   0:00.00 rcu_exp_gg_kthread_worker/0
17 root       20   0 0 0 0 0 S    0.0   0.0   0:00.00 rcu_exp_gg_kthread_worker
18 root       rt   0 0 0 0 0 S    0.0   0.0   0:00.01 migration/0
19 root       20   0 0 0 0 0 S    0.0   0.0   0:00.00 kprobe-optimizer
20 root      -SI   0 0 0 0 0 S    0.0   0.0   0:00.00 idle_inject/0
21 root       20   0 0 0 0 0 S    0.0   0.0   0:00.00 cpuhp/0
22 root       20   0 0 0 0 0 S    0.0   0.0   0:00.00 cpuhp/1
23 root      -SI   0 0 0 0 0 S    0.0   0.0   0:00.00 idle_inject/1
24 root       rt   0 0 0 0 0 S    0.0   0.0   0:00.01 migration/1
25 root       20   0 0 0 0 0 S    0.0   0.0   0:00.00 ksoftirqd/1
27 root       0 -20 0 0 0 0 I    0.0   0.0   0:00.00 kworker/1:0H-kblockd
28 root       20   0 0 0 0 0 S    0.0   0.0   0:00.00 kdevtmpfs
29 root       0 -20 0 0 0 0 I    0.0   0.0   0:00.00 kworker/R-inet_frag_wq
30 root       20   0 0 0 0 0 I    0.0   0.0   0:00.00 rcu_tasks_kthread
31 root       20   0 0 0 0 0 I    0.0   0.0   0:00.00 rcu_tasks_rude_kthread
32 root       20   0 0 0 0 0 S    0.0   0.0   0:00.00 kauditd
33 root       20   0 0 0 0 0 S    0.0   0.0   0:00.00 khungtaskd
35 root       20   0 0 0 0 0 S    0.0   0.0   0:00.00 oom_reaper
37 root       0 -20 0 0 0 0 I    0.0   0.0   0:00.00 kworker/R-writeback
38 root       20   0 0 0 0 0 S    0.0   0.0   0:00.20 kcompactd0
39 root       25   5 0 0 0 0 S    0.0   0.0   0:00.00 ksmd
40 root       39  15 0 0 0 0 S    0.0   0.0   0:00.00 khugepaged
41 root       0 -20 0 0 0 0 I    0.0   0.0   0:00.00 kworker/R-kblockd
```

Task 4 - Secure monitoring logs by restricting access to a dedicated user.

🚩 Objective

1. Create a dedicated user for monitoring operations.

- 2 . Create monitoring directory: /opt/container-monitor
- 3 . Assign ownership of monitoring folder to created user.
- 4 . Provide full access to monitoring user.
5. Restrict access for other users.
6. Verify access control.

```
MINGW32/c/Users/Others/downloads/docker-project/Task-1
root@ip-172-31-39-215:~# adduser monitoruser
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for monitoruser
Enter the new value, or press ENTER for the default
Full Name []: siva prasad singu
Room Number []: 402
Work Phone []: 630246170
Home Phone []: 8897001016
Other []:
Is the information correct? [Y/n] y
root@ip-172-31-39-215:~# chown -R monitoruser:monitoruser /opt/container-monitor
root@ip-172-31-39-215:~# chmod -R 700 /opt/container-monitor
root@ip-172-31-39-215:~# ls -ld /opt/container-monitor
drwx----- 3 monitoruser monitoruser 4096 Jun 19 01:48 /opt/container-monitor
root@ip-172-31-39-215:~# cd /opt/container-monitor
root@ip-172-31-39-215:/opt/container-monitor# su - monitoruser
monitoruser@ip-172-31-39-215:~$
monitoruser@ip-172-31-39-215:~$ cd /opt/container-monitor
ls
logs monitor.sh
monitoruser@ip-172-31-39-215:/opt/container-monitor$ crontab -u monitoruser -e
no crontab for monitoruser - using an empty one
Select an editor. To change later, run select-editor again.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# h dom mon dow   command
crontab: installing new crontab
monitoruser@ip-172-31-39-215:/opt/container-monitor$ crontab -u monitoruser -l
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# h dom mon dow   command
```

```
MINGW32/c/Users/Others/downloads/docker-project/Task-1
2026-06-19 03:15:01 | CPU: 0.00% | MEM: 3.5Mib / 908.7Mib
2026-06-19 03:15:01 | CPU: | MEM:
2026-06-19 03:15:01 | CPU: 0.00% | MEM: 3.5Mib / 908.7Mib
monitoruser@ip-172-31-39-215:/opt/container-monitor$ ls -l /opt/container-monitor/logs/
total 8
-rwx----- 1 monitoruser monitoruser 4338 Jun 19 03:15 monitor.log
monitoruser@ip-172-31-39-215:/opt/container-monitor$ cd ..
-bash: cd: ..: No such file or directory
monitoruser@ip-172-31-39-215:/opt/container-monitor$ cd
monitoruser@ip-172-31-39-215:/opt/container-monitor$ sudo apt update
sudo: I'm sorry monitoruser, I'm afraid I can't do that
sudo: I'm sorry monitoruser, I'm afraid I can't do that
monitoruser@ip-172-31-39-215:~$ exit
logout
root@ip-172-31-39-215:/opt/container-monitor# cd
root@ip-172-31-39-215:~# sudo apt update
Get:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:4 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [257 kB]
Get:5 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [68.9 kB]
Get:6 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [49.4 kB]
Get:7 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64 c-n-f Metadata [4400 B]
Get:8 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [150 kB]
Get:9 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [47.0 kB]
Get:10 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [62.1 kB]
Get:11 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 c-n-f Metadata [3908 B]
Get:12 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [208 kB]
Get:13 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [36.1 kB]
Get:14 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:15 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [31.2 kB]
Get:16 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.1 kB]
Fetched 1256 kB in 0s (2680 kB/s)
17 packages can be upgraded. Run 'apt list --upgradable' to see them.
ufw is already the newest version (0.36.2-9ubuntu1).
ufw set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 17
root@ip-172-31-39-215:~# sudo ufw status
Status: inactive
root@ip-172-31-39-215:~# sudo ufw allow ssh
Rules updated
Rules updated (v6)
root@ip-172-31-39-215:~# sudo ufw allow from YOUR_IP to any port 22
ERROR: Bad source address
root@ip-172-31-39-215:~# client_loop: send disconnect: Connection reset by peer

Others@VIJAYKUMAR_MINGW32 ~Downloads
$ ssh -i "ssp.pem" ubuntu@ec2-13-207-4-89.ap-south-1.compute.amazonaws.com
Welcome to Ubuntu 20.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro
```

Task 5 - Firewall Configuration (UFW)

🚩 Objective

Secure server by restricting SSH access.

🔧 Commands Executed

```
sudo ufw enable
```

```
sudo ufw status numbered
```

```
sudo ufw delete 1
```

```
sudo ufw delete 2
```

```
sudo ufw allow from 13.207.4.89 to any port 22
```

```
sudo ufw allow 80
```

```
sudo ufw allow 8000
```

🗝️ Result

```
MINGW32/c/Users/Others/downloads/docker-project/Task-1
$ ssh -i "ssp.pem" ubuntu@ec2-13-207-4-89.ap-south-1.compute.amazonaws.com
welcome to ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Fri Jun 19 03:27:30 UTC 2026

System load: 0.0          Temperature: -273.1 C
Usage of /:  42.0% of 6.6GB Processes:      141
Memory usage: 40%        Users logged in: 0
Swap usage:  0%          IPv4 address for ens5: 172.31.39.215

 * Ubuntu Pro delivers the most comprehensive open source security and
  compliance features.
  https://ubuntu.com/aws/pro

Expanded Security Maintenance for Applications is not enabled.

14 updates can be applied immediately.
14 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Fri Jun 19 02:59:18 2026 from 106.221.187.179
ubuntu@ip-172-31-39-215:~$ sudo -i
root@ip-172-31-39-215:~# curl ifconfig.me
13.207.4.89root@ip-172-31-39-215#sudo ufw allow from 13.207.4.89 to any port 22any port 22
Rules updated
root@ip-172-31-39-215:~# sudo ufw status numbered
Status: inactive
root@ip-172-31-39-215:~# sudo ufw allow from 13.207.4.89 to any port 22
Skipping adding existing rule
root@ip-172-31-39-215:~# sudo ufw status numbered
Status: inactive
root@ip-172-31-39-215:~# sudo ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y/n)? y
Firewall is active and enabled on system startup
root@ip-172-31-39-215:~# sudo ufw status numbered
Status: active

      To Action From
      --
[ 1] 22/tcp ALLOW IN Anywhere
[ 2] 22 ALLOW IN 13.207.4.89
[ 3] 22/tcp (v6) ALLOW IN Anywhere (v6)

root@ip-172-31-39-215:~# sudo ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y/n)? y
Firewall is active and enabled on system startup
```

```
MINGW32/c/Users/Others/downloads/docker-project/task-1
Rule deleted
root@ip-172-31-39-215:~# sudo ufw status numbered
Status: active

To Action From
--
[ 1] 22 ALLOW IN 13.207.4.89
[ 2] 22/tcp (v6) ALLOW IN Anywhere (v6)

root@ip-172-31-39-215:~# sudo ufw delete 2
Deleting:
allow 22/tcp
Proceed with operation (y/n)? y
Rule deleted (v6)
root@ip-172-31-39-215:~# sudo ufw status
Status: active

To Action From
--
22 ALLOW 13.207.4.89

root@ip-172-31-39-215:~# sudo ufw allow 80
Rule added
Rule added (v6)
Rule added (v6)
root@ip-172-31-39-215:~# sudo ufw status numbered
Status: active

To Action From
--
[ 1] 22 ALLOW IN 13.207.4.89
[ 2] 80 ALLOW IN Anywhere
[ 3] 8000 ALLOW IN Anywhere
[ 4] 80 (v6) ALLOW IN Anywhere (v6)
[ 5] 8000 (v6) ALLOW IN Anywhere (v6)

root@ip-172-31-39-215:~# client_loop: send disconnect: Connection reset by peer

Others@V13AYKUMAR MINGW32 ~/downloads
$ ssh -i "ssp.pem" ubuntu@ec2-13-207-4-89.ap-south-1.compute.amazonaws.com

Others@V13AYKUMAR MINGW32 ~/downloads
$ git clone https://github.com/sivaprasad112/docker-project.git
Cloning into 'docker-project'...
warning: You appear to have cloned an empty repository.

Others@V13AYKUMAR MINGW32 ~/downloads
$ cd docker-project

Others@V13AYKUMAR MINGW32 ~/downloads/docker-project (main)
$ mkdir Task-1
```

- SSH restricted to specific IP
- Ports 80 and 8000 open